NINE MUTUALLY ORTHOGONAL CYCLIC 9-CYCLE SYSTEMS OF K_{19}

Abstract. Burgess, Cavenagh, and Pike reported eight mutually orthogonal cyclic 9-cycle systems of K_{19} and stated that they did not know whether a larger family exists. We give an explicit family of nine. Consequently,

$$9 \le \mu'(9,19) \le 16.$$

An ℓ-cycle system of K_n is a partition of $E(K_n)$ into ℓ-cycles. Two such systems are *orthogonal* if every cycle in one shares at most one edge with every cycle in the other. A system on vertex set $\mathbb{Z}_n$ is *cyclic* if it is invariant under translation. Let $\mu'(\ell,n)$ be the maximum size of a pairwise orthogonal family of cyclic ℓ-cycle systems of K_n.

Burgess, Cavenagh, and Pike found a family of size eight for $(\ell,n) = (9,19)$ and left open whether a larger family exists [1, Appendix, Table 1]. We answer this question affirmatively.

Theorem 1. *There exist nine mutually orthogonal cyclic 9-cycle systems of K_{19}. In particular, $\mu'(9,19) \ge 9$.*

For the proof, let C be a cycle in $\mathbb{Z}_{19}$ and, for $t \in \mathbb{Z}_{19}$, let $C+t$ be obtained by adding t to every vertex. Set

$$\mathcal{O}(C) = \{C+t : t \in \mathbb{Z}_{19}\}.$$

The *difference* of an edge $\{x,y\}$ is the unique $d \in \{1,\dots,9\}$ such that $y - x \equiv \pm d \pmod{19}$.

Lemma 2. *Let C and D be 9-cycles, each containing exactly one edge of every difference $d \in \{1,\dots,9\}$. Orient each difference-d edge in the positive-d direction, and let a_d and b_d be its tail in C and D, respectively. Then $\mathcal{O}(C)$ and $\mathcal{O}(D)$ are orthogonal if and only if the residues*

$$a_d - b_d, \qquad 1 \le d \le 9,$$

are pairwise distinct modulo 19.

Proof. For $x, y \in \mathbb{Z}_{19}$, the difference-d edges of $C+x$ and $D+y$ coincide if and only if

$$a_d + x = b_d + y,$$

or equivalently $a_d - b_d = y - x$. Thus the number of common edges of $C+x$ and $D+y$ is the multiplicity of $y-x$ among the nine residues $a_d - b_d$. The result follows. $\square$

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Proof of Theorem 1. All arithmetic is in $\mathbb{Z}_{19}$. Let

$$C = (0, 1, 3, 18, 5, 13, 4, 9, 12) \qquad \text{and} \qquad \mathcal{F}_0 = \mathcal{O}(C).$$

Including the closing edge, the signed consecutive differences of C are

$$(1, 2, -4, 6, 8, -9, 5, 3, 7),$$

so C contains exactly one edge of every difference $1, \ldots, 9$. For each d, translating its difference-d edge gives all 19 edges of that difference class exactly once. Since $9 \cdot 19 = \binom{19}{2}$, $\mathcal{F}_0$ is a cyclic 9-cycle system of K_{19}.

The element $4 \in \mathbb{Z}_{19}^{\times}$ has order 9: one has $4^9 \equiv 1 \pmod{19}$, whereas $4 \not\equiv 1$ and $4^3 \equiv 7 \not\equiv 1 \pmod{19}$. For $0 \le r \le 8$, define

$$\mathcal{F}_r = 4^r \mathcal{F}_0.$$

Multiplication by 4^r permutes $\mathbb{Z}_{19}$, and hence

$$4^r \mathcal{F}_0 = \{4^r C + 4^r t : t \in \mathbb{Z}_{19}\} = \mathcal{O}(4^r C).$$

Thus every $\mathcal{F}_r$ is a cyclic 9-cycle system.

For $\mathcal{F}_0$, the tails of the positively oriented difference-d edges, indexed by $d = 1, \ldots, 9$, are

$$a^{(0)} = (0, 1, 9, 18, 4, 18, 12, 5, 4).$$

Let $a^{(r)}$ be the corresponding tail vector for $4^r C$. Direct calculation gives

$r \backslash d$	1	2	3	4	5	6	7	8	9
1	3	6	16	18	8	4	2	1	13
2	16	1	12	11	5	8	9	6	2
3	4	14	3	6	2	9	12	5	11
4	11	12	2	15	6	5	14	0	4

$$\left(a_d^{(0)} - a_d^{(r)}\right)_{d=1}^{9}.$$

Each row contains nine distinct residues modulo 19. By Lemma 2,

$$\mathcal{F}_0 \perp \mathcal{F}_r \qquad (1 \le r \le 4).$$

Multiplication by a nonzero residue preserves edge intersections. Hence any pair $\mathcal{F}_i, \mathcal{F}_j$ is carried, after multiplication by 4^{-i}, to $\mathcal{F}_0, \mathcal{F}_{j-i}$, with exponents modulo 9. Orthogonality is symmetric, so the cases r and $9-r$ are equivalent. The four checks above therefore cover every pair, and $\mathcal{F}_0, \ldots, \mathcal{F}_8$ are mutually orthogonal. $\square$

By Theorem 1 and the general bound $\mu'(\ell, n) \le n-3$ from [1, Lemma 6.4],

$$9 \le \mu'(9, 19) \le 16.$$

The construction does not rule out a family of size ten or larger.

References

[1] A. C. Burgess, N. J. Cavenagh, and D. A. Pike, *Mutually orthogonal cycle systems*, Ars Math. Contemp. **23** (2023), no. 2, Paper No. P2.05, 20 pp., doi:10.26493/1855-3974.2692.86d.